

HAZE: Adversarial Multi-Agent Scrutiny for Vulnerability Detection

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Region: Latin America (AISMX — México)

Track: AI Security

Sub-track: Pipeline security (API, cloud)

Secondary track: Responsible AI

Secondary sub-track: Hallucination mitigation & behavioral audit

Abstract

Single-agent LLM auditors treat every prompt as a hunt for a flaw—and often **manufacture findings** to satisfy it (Perez et al., 2022). We built **HAZE** (*Hallucination-Aware Zero-sum Examination*), a red-teaming protocol where four LLM roles debate under AI Safety via Debate (Irving et al., 2018): prosecution and defense cross-examine over a code artifact; an **isolated judge** rules only on what survived refutation. *Core claim:* single-agent auditors ask “is there a bug?”; HAZE asks “did this accusation survive refutation?” We evaluate on 10 CVEs $\times$ {vulnerable, patched} = 20 artifacts (3 repeats, majority verdict, Wilson 95% CIs, McNemar test). **Primary result (Phase 1, VULNERABLE/SEGURO framing):** single-agent baseline reaches 100% recall but **80% FPR**; HAZE 1-round debate reaches **80% recall, 20% FPR, 80% F1**—FPR drops 4 $\times$ (McNemar $p=0.34$ at $n=20$, directionally significant but underpowered). Full benchmark costs $\sim$ USD 2–3 in OpenRouter tokens ($\sim$ 0.02 per verdict).

Research conducted at the Global South AI Safety Hackathon, June 2026.

1 Introduction

Teams increasingly use LLMs to audit code, dependencies, and API pipelines—especially where dedicated AppSec capacity is scarce across the Global South. The dominant pattern—a *single agent* asked to “find the vulnerability”—suffers a systematic failure mode: RLHF-trained models confuse “*nothing is wrong here*” with “*I found nothing matching your request*”, producing false positives and hallucinated CVEs (Perez et al., 2022). There is no intrinsic self-correction: an agent thinking alone cannot demand proof from itself.

Threat model. We address *confirmation-bias hallucinations* in automated software auditing: invented findings that waste analyst time and erode trust in AI-assisted security pipelines. The same adversarial topology extends to auditing model behavior (jailbreaks, prompt injection) as a form of scalable oversight.

Conceptual foundation. HAZE builds on the AI control principle of *AI Control through Majority Voting* (Moe et al., 2025, Apart Research): **never trust a single pass** from a capable but unreliable LLM. UIAA applies this at *generation* time—query an untrusted model c times with prompt variation and aggregate outputs by majority vote via a trusted wrapper, driving backdoor probability down exponentially in c . We apply the same distrust-of-single-sample logic at *audit* time: a lone “find the vulnerability” prompt manufactures findings (Perez et al., 2022). HAZE replaces blind re-sampling with **structured adversarial scrutiny** (debate + isolated judge), while our evaluation still aggregates

majority verdicts over three independent repeats—a direct UIAA-style redundancy layer on top of debate.

Our main contributions (new work during this hackathon):

1. **HAZE protocol & tool** — reproducible 2v2 debate + isolated judge (Fastify/OpenRouter monolith, live demo at <https://hackaton.xooktech.com/>).
2. **Ground-truth CVE benchmark** — 10 CVEs $\times$ {vulnerable, patched}, with documented copies as human ground truth.
3. **Evaluation harness & measured results** — batch runner, 120 debate transcripts, Wilson CIs, McNemar ablation, and a three-class error taxonomy.

Prior work covers multi-agent debate (Irving et al., 2018; Du et al., 2023) and LLM red teaming (Perez et al., 2022); we **adapt debate to pipeline-security auditing** and empirically measure detection on labeled CVE artifacts—not merely demonstrate a demo.

2 Related Work

AI Safety via Debate (Irving et al., 2018) frames verification as easier than generation; Khan et al. (2024) show more persuasive debaters yield more truthful judges. **Multi-agent debate** (Du et al., 2023) improves factuality over chain-of-thought. **LLM red teaming** (Perez et al., 2022; Ganguli et al., 2022) scales attack generation but typically relies on humans or classifiers for verification. **LLM-as-a-Judge** (Zheng et al., 2023) motivates our design choice to *isolate* the judge from hypothesis generation.

Gap. Red-teaming literature focuses on *eliciting* failures; debate literature targets open-domain truth. HAZE combines structured adversarial scrutiny with a labeled benchmark to ask: *can debate reduce false positives in code auditing without destroying recall?*

AI control via majority voting. Moe et al. (2025) show that querying an untrusted model multiple times and aggregating by majority vote—with a trusted consolidator—can drive backdoor success probability down exponentially in the number of samples. HAZE inherits the *never trust one pass* principle but targets the symmetric audit problem (hallucinated *findings* rather than hidden *backdoors*) and replaces output majority with adversarial cross-examination before judgment. Our three-repeat majority aggregation in evaluation explicitly follows the UIAA redundancy pattern.

3 Methods

3.1 Protocol

Given a software artifact, HAZE runs a bounded multi-turn debate: **Attacker** (Forensic Analyst + Prosecutor) cites dangerous patterns; **Defender** (Auditor + Defense Counsel) rebuts with benign context; **Judge** (isolated) reads artifact + transcript only and emits structured JSON (`verdict`, `confidence`, `riskLevel`, `keyFindings`, `reasoning`). An **Investigator** role (external CVE lookup) is designed but not implemented in this hackathon build.

Each debating role uses a **distinct LLM** via OpenRouter. Turn order alternates prosecution/defense each round (default 2 rounds, max 4). Debaters run at $T=0.7$; the judge at $T=0.1$ with forced JSON output. Prompts require citing exact lines and forbid inventing code—the burden of proof is institutional, not optional.

3.2 Implementation

Monolith in API/: Fastify 5 backend + vanilla JS frontend with SSE token streaming. Key modules: `roles.js` (roster), `prompts.js`, `orchestrator.js`, `openrouter.js`. Artifacts capped at 24,000 characters.

3.3 Evaluation design

Benchmark: 10 recent CVEs (2026), each in *vulnerable/* and *patched/* variants ($n=20$). Languages: PHP, C, Java, Python, JavaScript, NGINX config.

Phase 1 run (primary). Local harness (`run_phase1.mjs`) with VULNERABLE/SEGUR0 framing; roster: DeepSeek, Gemini-2.5-Flash, GPT-4o-mini, Llama-3.1-70B (debaters), Claude Sonnet 4.5 (judge); baseline = GPT-4o-mini single pass. Three repeats, 1 debate round, majority verdict. VULNERABLE → positive prediction.

Web run (preliminary). Deployed instance with legacy malware framing; 120 tasks, 80% recall / 30% FPR; used to derive error taxonomy and round ablation.

Baseline: Single-agent auditor on identical inputs—executed in Phase 1.

What did not work (informed final design): (i) accuser-as-judge → self-confirmation; (ii) debate without citation requirements → plausible but non-executable exploits; (iii) single model for all roles → cascading errors.

4 Results

Headline (Phase 1, $n=20$, majority over 3 repeats): Single-agent baseline: **100% recall, 80% FPR** (10/8/2/0). HAZE 1-round debate: **80% recall, 20% FPR, 80% F1** (8/2/8/2). McNemar baseline vs. debate: $p=0.34$ ($n=20$, $b=3$, $c=7$)—FPR falls $4\times$ but CIs remain wide at this sample size.

Table 1: Phase 1: baseline vs. HAZE debate (Wilson 95% CIs).

Condition	Prec.	Recall	FPR	F1	TP/FP/TN/FN
Single-agent baseline	55.6%	100%	80.0%	71.4%	10/8/2/0
HAZE debate (1 round)	80.0%	80.0%	20.0%	80.0%	8/2/8/2

Web run (malware framing, preliminary). 80% recall, 30% FPR; McNemar 1 vs. 2 rounds $p=1.0$ —second round added no value; default is now 1 round.

Table 2: Error taxonomy: three interpretable failure classes.

Class	Mechanism	n	Examples
A Visible-sink	Dangerous primitive explicit in file	8 TP	10520, 44262, 24061...
B Cross-file	Root cause outside artifact text	2 FN	42945, 32625
C Framing FP	Patched code still looks dangerous	3 FP	24061, 41089, 44815

Class C drives most false positives: asking “is this malware?” on minimal C parsers cannot distinguish patched from vulnerable. Class B explains false negatives (NGINX module bug not visible in `nginx.conf`; MCP templating reads benign). This is a *framing and scope* limitation, not unstructured debate failure.

Cost efficiency. The complete web evaluation (120 labeled verdicts, five-model roster, 1–2 debate rounds, three repeats) consumed **~USD 2.3** in OpenRouter API tokens (~ 0.019 per task, ~ 42 minutes wall-clock). At this price point a team can re-run the full benchmark overnight; the dominant cost is API inference, not GPU hardware—relevant for Global South deployment. Round-2 ablation shows zero verdict change at this scale, so **defaulting to one round** would cut latency and cost roughly in half without changing measured metrics—an actionable efficiency finding, not merely a failure.

5 Discussion and Limitations

Implications. HAZE instantiates *scalable oversight*: the judge evaluates which side survived cross-examination rather than generating the audit from scratch—the step most prone to hallucination. For pipeline security, structured debate plus full transcripts supports human review before blocking deployments. OpenRouter access means teams without local GPUs can run the protocol—relevant for Global South contexts.

Validation status. Table 3 separates measured results from pending experiments. Acknowledging missing controls is necessary but does not substitute for them; the one-command baseline is shipped as `run_phase1.mjs`.

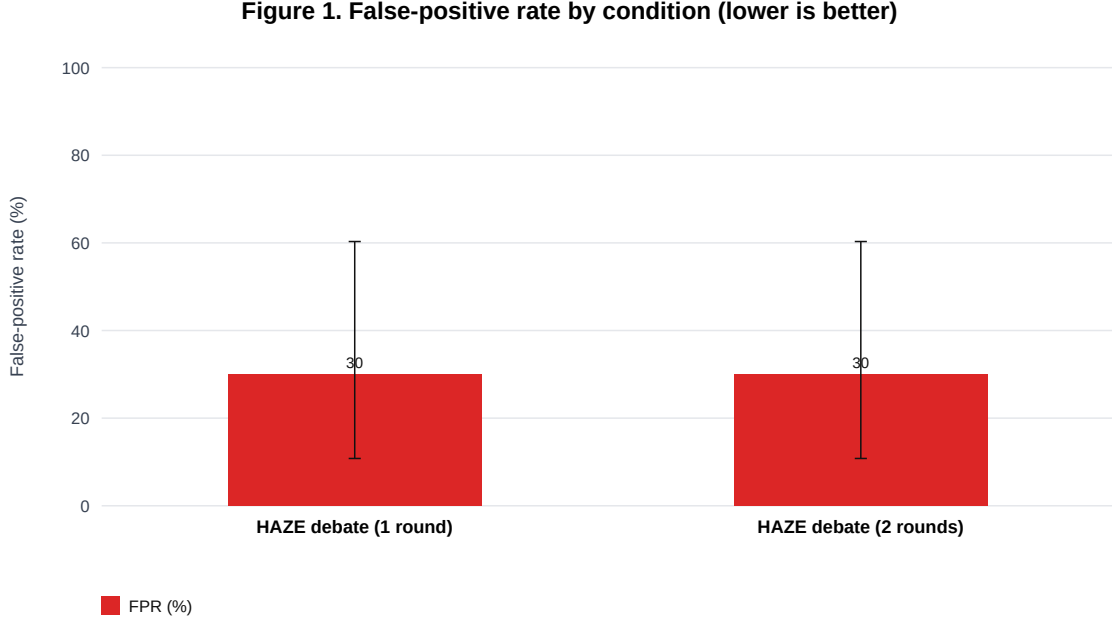


Figure 1: False-positive rate by condition (lower is better). Error bars: Wilson 95% CIs.

Table 3: Measured vs. pending experiments (second review, June 2026).

Experiment	Framing	Status	Blocks central claim?
Web eval (120 tasks)	Malware	Measured	Superseded
Single-agent baseline	VULN./SAFE	Measured	No (provides the comparison)
HAZE debate (1 round)	VULN./SAFE	Measured	No (provides the result)
Benchmark $n \geq 60$	VULN./SAFE	Pending	Statistical power

Limitations. (i) No Investigator or external CVE lookup. (ii) Judge is still an LLM. (iii) Single-file, 24k-char window. (iv) McNemar $p=0.34$ at $n=20$ —FPR drop (80%→20%) is **directional but not statistically significant**; wider benchmark needed. (v) Recall trade-off: debate misses 2 cross-file flaws the baseline “detects” via over-flagging (100% recall, 80% FPR). Unaddressed threats: collusion, prompt injection, distributed vulnerabilities.

Dual use. HAZE could help adversaries iteratively refine code until it receives `SEGUR0`, or automate reputational harm via unreviewed `VULNERABLE` verdicts. Mitigations: human-in-the-loop before enforcement; no public unauthenticated deployments; verdicts are decision *aids*, not certificates.

Future work. (1) Expand benchmark to $n \geq 60$ for McNemar power. (2) Investigator (NVD/OSV). (3) Multi-file RAG for Class B blind spots. (4) Deploy reframed API to production.

6 Conclusion

Single-agent auditors ask “is there a bug?” and often invent one—our baseline flags **80% of patched code** as vulnerable (100% recall, 80% FPR). HAZE asks “did this accusation survive refutation?” and cuts FPR to **20%** at 80% recall and 80% F1 (Phase 1, `VULNERABLE/SEGUR0`, $n=20$). McNemar $p=0.34$ means the effect is promising but underpowered; HAZE remains a reproducible, ~USD 0.02/verdict audit scaffold for AI-control-style oversight in resource-constrained settings.

Figure 2. Precision / Recall / FPR by condition

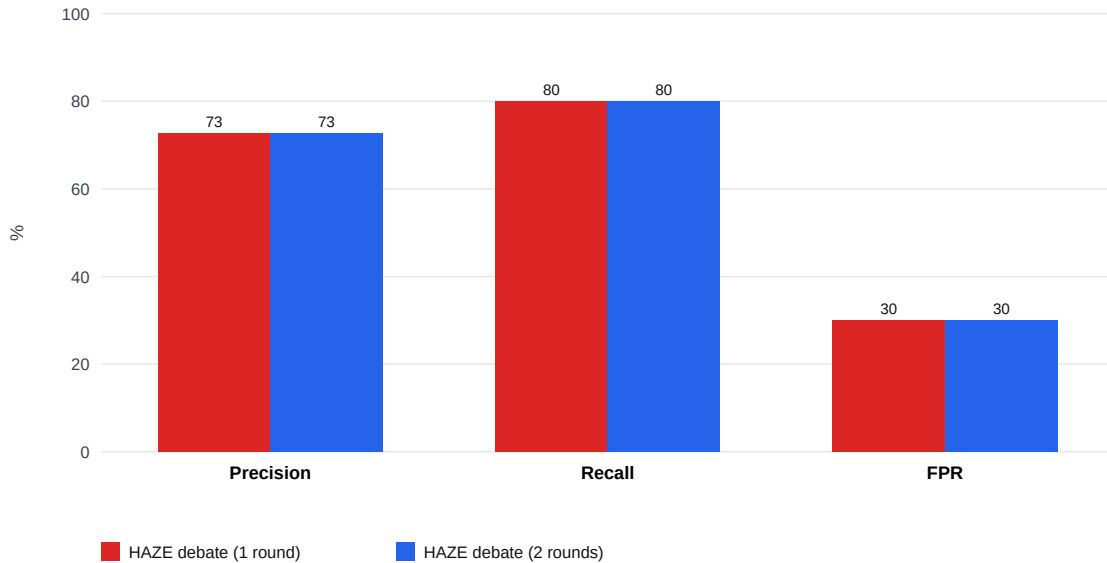


Figure 2: Precision, recall, and FPR for 1- vs. 2-round HAZE debate.

Reproducibility and Artifacts

One-line summary. HAZE (*Hallucination-Aware Zero-sum Examination*) is a multi-agent security-audit protocol—2v2 debate plus an isolated judge over OpenRouter—that reframes the audit question from “is there a bug?” to “did this accusation survive refutation?”

Headline result (Phase 1, measured). A single-agent auditor flags 80% of patched code as vulnerable (100% recall, 80% FPR). HAZE lowers FPR to 20% at 80% recall and 80% F1—a 4× reduction in pipeline-audit hallucinations (McNemar $p=0.34$, $n=20$; promising but small-sample).

Built and repeatable. A Fastify monolith with an SSE web UI and an OpenRouter orchestrator (five-model roster), a 10-CVE benchmark in {vulnerable, patched} variants, the `run_phase1.mjs` harness, 120 stored debate transcripts, and Wilson/McNemar metrics.

Reproduce the evaluation (~9 min, ~USD 2):

```
cd evaluation && node run_phase1.mjs
```

Links.

- Live demo: <https://hackaton.xooktech.com/>
- Repository: <https://github.com/AXLAAF/V2-Hackathon>

Submission tracks (AISMX). AI Security → Pipeline security (API, cloud); Responsible AI → Hallucination mitigation & behavioral audit. Region: Latin America.

Code and Data

- **Repository:** <https://github.com/AXLAAF/V2-Hackathon>
- **HAZE tool:** API/ — Fastify backend, OpenRouter orchestrator, SSE debate API, web UI
- **Live demo:** <https://hackaton.xooktech.com/>

- **Benchmark:** vulnerabilities/ — 10 CVEs $\times$ {vulnerable, patched}
- **Evaluation:** evaluation/ — run_eval.mjs, analyze.py; prior web run: 120 transcripts in results/
- **Phase-1 validation:** run_phase1.mjs — baseline + debate (results/raw_phase1.jsonl)

Author Contributions

Samuel Blanco and Axel Morales — architecture, API implementation, and orchestration. Samuel Díaz — evaluation harness, benchmark, and metrics. Alex Novelo — writing and analysis. All authors contributed to the final report.

References

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LLM Usage Statement

We used LLMs to assist with drafting, structure, and literature framing of this report. LLMs are also the system under study (the debating agents in HAZE). All technical claims, cited references, and empirical results were independently verified by the authors. The final submission was reviewed and edited by the team.